

```
MariaDB [(none)]> create database CompanyDB;
Query OK, 1 row affected (0.034 sec)
```

```
MariaDB [(none)]> use CompanyDB;
Database changed
```

```
MariaDB [CompanyDB]> create table Department(deptNo varchar(10) primary key,
deptName varchar(50) NOT NULL, mngrEmpNO varchar(10));
Query OK, 0 rows affected (0.059 sec)
```

```
MariaDB [CompanyDB]> create table Employee(empNo varchar(10) primary key, FName
varchar(20) Not Null, Lname varchar(20) Not Null, Address varchar(100), DOB Date,
Sex char(1) Not Null, check(sex IN('M', 'F')));
Query OK, 0 rows affected (0.035 sec)
```

```
MariaDB [CompanyDB]> create table Employee(empNo varchar(10) primary key, FName
varchar(20) Not Null, Lname varchar(20) Not Null, Address varchar(100), DOB Date,
Sex char(1) Not Null check(sex IN('M', 'F')), deptNo varchar(10), Position
varchar(50) check(Position IN('Manager', 'Team Leader', 'Analyst', 'Software
Developer')), FOREIGN KEY(deptNo) References Department(deptNO) ON Delete set
Null);
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [CompanyDB]> Alter table Department Add Constraint FK_dept_manager FOREIGN
KEY(MngrEmpNO) REFERENCES Employee(EmpNO) ON Delete set Null;
Query OK, 0 rows affected (0.044 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [CompanyDB]> create table Project(projNO Varchar(10) Primary Key, ProjName
varchar(100) Not Null, deptNo varchar(10), FOREIGN KEY(deptNo) REFERENCES
Department(deptNO) ON Delete CASCADE);
Query OK, 0 rows affected (0.030 sec)
```

```
MariaDB [CompanyDB]> Create table WorksON(empNO varchar(10), projNo varchar(10),
DateWorked DATE, HoursWorked INT check(HoursWorked BETWEEN 0 and 40), PRIMARY
KEY(empNO, ProjNO, Dateworked), FOREIGN KEY(empNo) REFERENCES Employee(empNO) ON
DELETE CASCADE, FOREIGN KEY(projNo) REFERENCES project(ProjNO) ON DELETE CASCADE);
Query OK, 0 rows affected (0.034 sec)
```

```
MariaDB [CompanyDB]> insert into department(deptNO, deptName, mngrEmpNo)
values('D01', 'IT', Null),('D02', 'Planning', Null),('D03', 'Finance', Null),
('D04', 'HR', Null), ('D05', 'Marketing', Null), ('D06', 'Operations', Null),
('D07', 'Research', Null);
Query OK, 7 rows affected (0.027 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

```
MariaDB [CompanyDB]> select * FROM department;
```

deptNo	deptName	mngrEmpNO
D01	IT	NULL

D02	Planning	NULL	
D03	Finance	NULL	
D04	HR	NULL	
D05	Marketing	NULL	
D06	Operations	NULL	
D07	Research	NULL	

+-----+-----+-----+-----+

7 rows in set (0.013 sec)

MariaDB [CompanyDB]> insert into employee(empNo, FName, LName, Address, DOB, Sex, deptNo, Position) Values

-> ('E001', 'John', 'Kamau', 'Nairobi', '1985-03-15', 'M', 'D01', 'Manager'),
('E002', 'Mary', 'Achieng', 'Kisumu', '1992-07-21', 'F', 'D02', 'Manager'), ('E003',
'Antony', 'Ebo', 'Accra', '1995-11-10', 'M', 'D01', 'Software Developer'), ('E004',
'Frank', 'Atta', 'Kumasi', '1998-03-09', 'F', 'D03', 'Analyst'), ('E005', 'James',
'Mwangi', 'Mobasa', '1988-09-14', 'M', 'D04', 'Manager'), ('E006', 'Akosua',
'Faith', 'Dome', '2001-01-30', 'F', 'D05', 'Team Leader'), ('E007', 'Peter',
'Kiptoo', 'Kitale', '1997-12-11', 'M', 'D06', 'Analyst');

Query OK, 7 rows affected (0.047 sec)

Records: 7 Duplicates: 0 Warnings: 0

MariaDB [CompanyDB]> select * from Employee;

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+

empNo	FName	Lname	Address	DOB	Sex	deptNo	Position
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+-----+-----+-----+-----+-----+-----+-----+-----+-----+

+-----+

E001	John	Kamau	Nairobi	1985-03-15	M	D01	Manager
------	------	-------	---------	------------	---	-----	---------

+-----+

E002	Mary	Achieng	Kisumu	1992-07-21	F	D02	Manager
------	------	---------	--------	------------	---	-----	---------

+-----+

E003	Antony	Ebo	Accra	1995-11-10	M	D01	Software Developer
------	--------	-----	-------	------------	---	-----	--------------------

+-----+

E004	Frank	Atta	Kumasi	1998-03-09	F	D03	Analyst
------	-------	------	--------	------------	---	-----	---------

+-----+

E005	James	Mwangi	Mobasa	1988-09-14	M	D04	Manager
------	-------	--------	--------	------------	---	-----	---------

+-----+

E006	Akosua	Faith	Dome	2001-01-30	F	D05	Team Leader
------	--------	-------	------	------------	---	-----	-------------

+-----+

E007	Peter	Kiptoo	Kitale	1997-12-11	M	D06	Analyst
------	-------	--------	--------	------------	---	-----	---------

+-----+-----+-----+-----+-----+-----+-----+-----+-----+

+-----+

7 rows in set (0.027 sec)

MariaDB [CompanyDB]> update department set mngrEmpNO = 'E001' where deptNO = 'D01';

Query OK, 1 row affected (0.029 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```

MariaDB [CompanyDB]> update department set mngrEmpNO = 'E002' where deptNO = 'D02';
MariaDB [CompanyDB]> update department set mngrEmpNO = 'E004' where deptNO = 'D03';
MariaDB [CompanyDB]> update department set mngrEmpNO = 'E005' where deptNO = 'D04';
MariaDB [CompanyDB]> update department set mngrEmpNO = 'E006' where deptNO = 'D05';
MariaDB [CompanyDB]> update department set mngrEmpNO = 'E007' where deptNO = 'D06';
MariaDB [CompanyDB]> update department set mngrEmpNO = 'E003' where deptNO = 'D07';
MariaDB [CompanyDB]> select * from department;

```

deptNo	deptName	mngrEmpNO
D01	IT	E001
D02	Planning	E002
D03	Finance	E004
D04	HR	E005
D05	Marketing	E006
D06	Operations	E007
D07	Research	E003

7 rows in set (0.001 sec)

```

MariaDB [CompanyDB]> insert into project(projNo, ProjName, deptNo) values ('P01',
'SCCS', 'D01'), ('P02', 'ERP', 'D01'), ('P03', 'Budget System', 'D03'), ('P04',
'Recruitment Portal', 'D04'), ('P05', 'Marketing Analytics', 'D05'), ('P06', 'Supply
Chain', 'D06'), ('P07', 'AI Research', 'D07');
Query OK, 7 rows affected (0.003 sec)
Records: 7 Duplicates: 0 Warnings: 0

```

```

MariaDB [CompanyDB]> select * from project;

```

projNO	ProjName	deptNo
P01	SCCS	D01
P02	ERP	D01
P03	Budget System	D03
P04	Recruitment Portal	D04
P05	Marketing Analytics	D05
P06	Supply Chain	D06
P07	AI Research	D07

7 rows in set (0.000 sec)

```

MariaDB [CompanyDB]> insert into workson(empNO, projNo, dateworked, hoursworked)
VALUES ('E001', 'P01', '2025-01-05',8), ('E002', 'P02', '2025-01-06',6), ('E003',
'P01', '2025-01-07',7), ('E004', 'P03', '2025-01-08',5), ('E005', 'P04',
'2025-01-09',8), ('E006', 'P05', '2025-01-10',6), ('E007', 'P06', '2025-01-11',7);
Query OK, 7 rows affected (0.003 sec)
Records: 7 Duplicates: 0 Warnings: 0

```

```

MariaDB [CompanyDB]> select * from Workson;

```

empNO	projNo	DateWorked	HoursWorked
E001	P01	2025-01-05	8
E002	P02	2025-01-06	6
E003	P01	2025-01-07	7
E004	P03	2025-01-08	5
E005	P04	2025-01-09	8
E006	P05	2025-01-10	6
E007	P06	2025-01-11	7

7 rows in set (0.000 sec)

VIEWS

a)

```
MariaDB [CompanyDB]> create VIEW EmpProjDetails As SELECT e.empNO, e.fName,
e.lName, p.projName, w.hoursworked FROM employee e JOIN Workson w ON e.empNo =
w.empNo JOIN project p ON p.projNo = w.projNo;
Query OK, 0 rows affected (0.011 sec)
```

```
MariaDB [CompanyDB]> select * from EmpProjDetails;
```

empNO	fName	lName	projName	hoursworked
E001	John	Kamau	SCCS	8
E002	Mary	Achieng	ERP	6
E003	Antony	Ebo	SCCS	7
E004	Frank	Atta	Budget System	5
E005	James	Mwangi	Recruitment Portal	8
E006	Akosua	Faith	Marketing Analytics	6
E007	Peter	Kiptoo	Supply Chain	7

7 rows in set (0.049 sec)

```
MariaDB [CompanyDB]> create VIEW EmpProject AS SELECT w.empNo, w.projNo,
SUM(w.hoursWorked) AS TotalHours FROM Employee e JOIN workson w ON e.empNo =
e.empNo JOIN Project p ON p.ProjNo = w.projNo GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.005 sec)
```

```
MariaDB [CompanyDB]> select * from EmpProject;
```

empNo	projNo	TotalHours
E001	P01	56
E002	P02	42
E003	P01	49
E004	P03	35

E005	P04	56
E006	P05	42
E007	P06	49

7 rows in set (0.012 sec)

b).

```
MariaDB [CompanyDB]> select projNo FROM EmpProject WHERE ProjNo = 'SCCS';
Empty set (0.005 sec)
```

c).

```
MariaDB [CompanyDB]> Select Count(projNo) FROM EmpProject Where empNo = 'E001';
```

Count(projNo)
1

1 row in set (0.007 sec)

d).

```
MariaDB [CompanyDB]> Select empNo, totalhours FROM empProject GROUP BY empNo;
```

empNo	totalhours
E001	56
E002	42
E003	49
E004	35
E005	56
E006	42
E007	49

7 rows in set (0.002 sec)

7.28

The materialized view stores the results of a query physically, so it must be updated whenever the Part table changes.

That is, if a new part has partcost that is greater than 1000, then its partNo is added to the view. Furthermore, if an expensive part is deleted, then it is necessary to check whether another contract for the same part still costs more than 1000 before removing it from the view.

The view can be maintained without accessing the part table during inserts or updates that make a part expensive because the new data is enough.

